

information-seeking questions are typically subjective, whereas information-seeking questions are typically objective.

Subjectivity is defined as the characteristic of being based on or influenced by personal feelings, tastes or opinions – it is more about the perceptions that exist in someone’s mind rather than the reality of the external world. Given this definition that subjectivity is a matter of an individual’s perception, one way of distinguishing between these two types of questions is to build a subjectivity classifier. This classifier takes as input a question sentence and classifies it as subjective or objective. Hence, this task is a binary short text classification problem.

There has been considerable work on subjectivity detection. Many of the earlier research papers have pointed out that the presence of adjectives is a strong indicator of subjectivity. The text is represented using a standard ‘bag-of-words’ model. In addition, we can capture ‘parts of speech’ tags for the text, and the sentiment contained in the text as features for subjectivity. Certain dependency relations in the text are also good indicators of subjectivity. For instance, dependency relations such as the adjective modifier relation ‘generous act’ or an adverbial modifier relation like ‘absolutely brilliant’ can often serve to indicate opinions or subjectivity. We can use the presence of these dependency relations as an additional feature for the objectivity classifier. We can also augment our representation using external lexicons to identify terms that can be potential indicators of subjectivity. Such lexicons include the MPQA subjectivity lexicon (available at https://mpqa.cs.pitt.edu/lexicons/subj_lexicon/), NRC emotion lexicon (<https://saifmohammad.com/WebPages/NRC-Emotion-Lexicon.htm>) and sentiment lexicons (<https://sentic.net/>). We can use the count of terms present in external subjectivity and emotion lexicons as an additional element in our feature vector. The constructed feature vector for the input is then fed to a standard supervised learning based classifier.

There are a number of works on subjectivity-detection using machine learning and natural language processing techniques. Of these, the most well known one is the OpinionFinder project, available at http://mpqa.cs.pitt.edu/opinionfinder/opinionfinder_2/. Since this is freely available software, we can use it to identify whether a question is subjective or not, based on whether it contains an opinion or not. If you want to train your own binary classifier for subjectivity detection, you can use the MPQA subjectivity corpus. This corpus consists of sentences marked as subjective or objective. However, note that these sentences are not questions, and hence it would be interesting to see how well a classifier

trained on non-question subjective sentences performs on detecting subjective questions. I leave this task as an experiment for our readers.

In addition to rhetorical questions, non-information-seeking questions can be intended to use inflammatory language against individuals or groups. They can also contain objectionable content/hate speech. Consider the following two questions.

Q5: Should Marxists be banned from Quora for their stupidity?

Q6: Are environmentalists the new Maoists?

These questions are intended to disparage people. Q5 targets followers of Karl Marx while question Q6 targets environmentalists. Both these questions express a negative sentiment against specific groups of people. Hence these are not information seeking in nature. However, we note that such questions that contain negative sentiments against groups or individuals can be classified correctly as subjective by the subjectivity classifier we discussed earlier. Hence most of the questions under this sub-category can be subsumed under the subjectivity classifier itself.

Another sub-category of non-information-seeking questions is of questions that express objectionable sexual content for shock value on topics such as incest, bestiality or paedophilia. We can build a classifier for such questions using an explicit dictionary for objectionable terms along with a ‘bag-of-words’ model. Another sub-category is hypothetical questions. These are questions that are based on unreal contexts and situations. Such questions are often in the form of a conditional statement such as, ‘If <some hypothetical condition>,’ followed by a question. For example, consider the following question.

Q7: If I don’t perform well in the JEE entrance exam, should I end my life?

We can use the presence of such conditionals along with the presence of certain modal verbs as features to build a classifier for hypothetical questions.

Now that we have separate classifiers for each of these categories, we can combine the decisions from all these classifiers and then create a final classification result. The simplest approach is to take the logical OR of all classifier decisions. Essentially, if any of the sub-categories gets flagged, we treat the question as an insincere question. We can also attempt to use a majority voting classifier operating on the results of these classifiers. So instead of just going by the final classification result, we can obtain the probability score for each of these decisions and combine them into a feature vector. This feature vector can be used to augment a content based neural representation of the input sentence, and the augmented representation can be fed to a neural network classifier.